

25W1 1: Inexact Halpern on Three VI Benchmarks Lower Bound, Forsaken, Polar Game)

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1 Introduction

This report summarizes tuning experiments for the *inexact Halpern* method on three VI benchmarks. We mainly control three parameters in code: (i) the stepsize η , (ii) damping via $\rho = \theta\eta$ (so $\alpha = 1 - \rho/\eta$), using the theoretical ρ unless the condition fails, and (iii) the inner Tseng-FBF accuracy through the coefficient c in N_k . While most experiments are run within the stability condition $0 < \rho < \eta < 1/L$, we also tested cases outside this range. In the **Forsaken** benchmark, additional tuning of the boundary and a was needed to satisfy the condition, and we also tried adjusting η to indirectly tune α and reduce limit-cycle behavior.

2 Experiments

2.1 Lower Bound

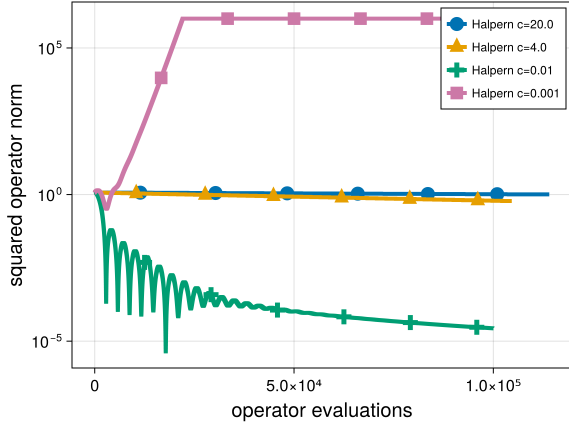
We consider the linear lower-bound instance with

$$\rho = -\frac{2b}{a^2 + b^2}, \quad L = \sqrt{a^2 + b^2},$$

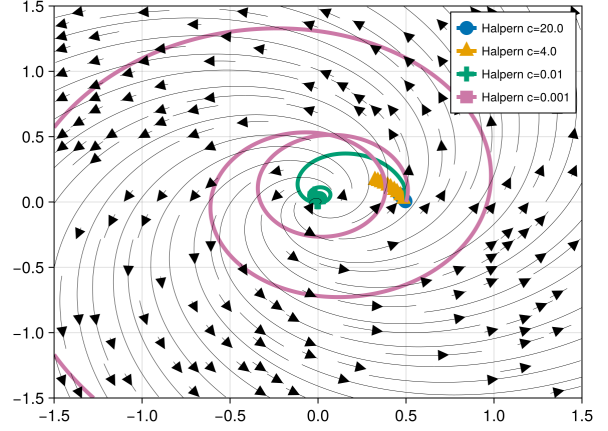
and first we use the problem-provided parameters $a^2 = 3.7$, $b = -1$ with $\rho = \frac{2}{4.7}$.

To satisfy the convergence condition for Halpern, the stepsize is set to $\eta = \frac{0.95}{L}$, which gives $\eta = 0.438202$ with $\rho = 0.425532$.

In practice, since the convergence condition was satisfied, Halpern converged for $c > 0.01$, whereas it diverged once c dropped below 0.01. Moreover, the smaller the value of c (while still above the stability threshold), the faster the convergence speed.



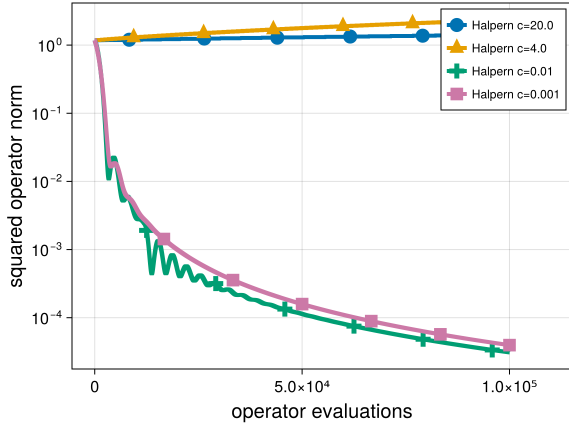
(a) Squared operator norm vs. operator evaluations



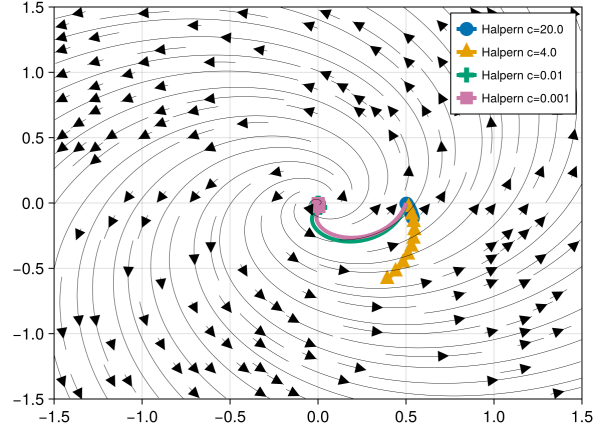
(b) Iterate trajectories over the vector field

Figure 1: Lower Bound ($\eta = \frac{0.95}{L}$). Converges for $c > 0.01$, with faster decay as c decreases.

However, when the convergence condition was not satisfied, all values of c diverged except when c was smaller than approximately 0.01.



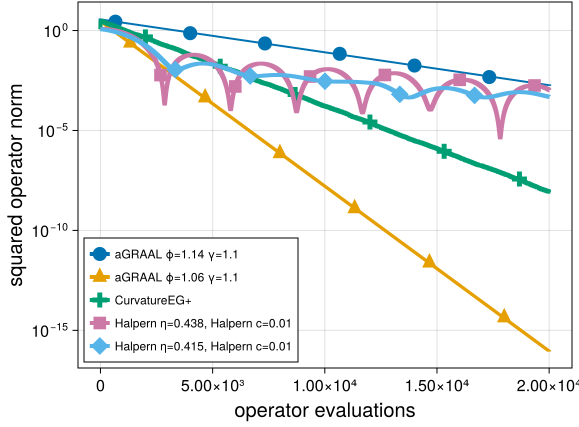
(a) Squared operator norm vs. operator evaluations



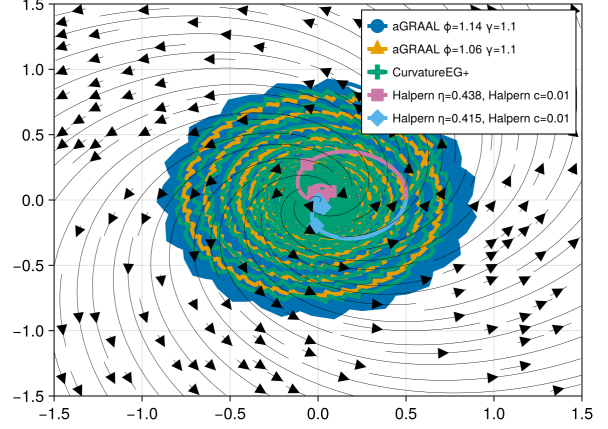
(b) Iterate trajectories over the vector field

Figure 2: Lower Bound ($\eta = \frac{0.9}{L}$). Stable only for very small c .

Interestingly, the inexact Halpern method showed the fastest initial decrease but became slower than the other methods in the later stages.



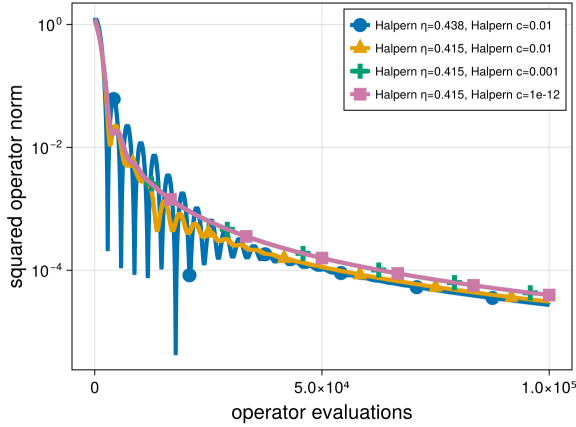
(a) Squared operator norm vs. operator evaluations



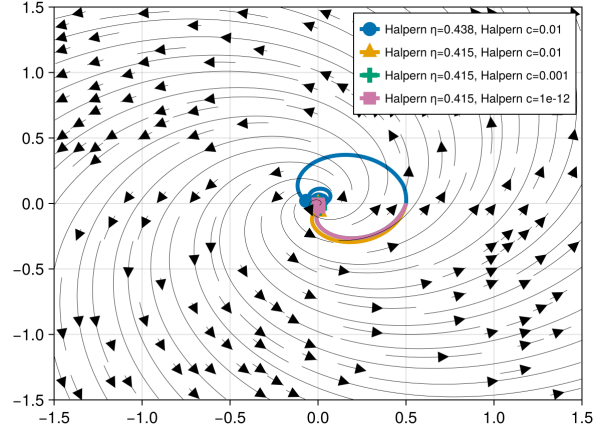
(b) Iterate trajectories over the vector field

Figure 3: Lower Bound. Comparison of $\eta = \frac{0.9}{L}$ and $\eta = \frac{0.95}{L}$ with $c = 0.01$.

Furthermore, in both cases Halpern used the same $c = 0.01$, yet the run at $\eta = 0.95/L$ (where the convergence condition was satisfied) was slightly faster than at $\eta = 0.9/L$ (where the condition was not satisfied). In the latter case, although the condition fails, we still observed convergence whenever c was chosen smaller than 0.01. One possible is that when the condition fails, the corresponding value of α becomes a small negative value -0.02503467559 , which can be regarded as a meaningless parameter.



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 4: Lower Bound. $\eta = \frac{0.9}{L}$ vs. $\eta = \frac{0.95}{L}$, both with $c = 0.01$.

Overall, the most interesting observation was that when the convergence condition was not satisfied ($\rho > \eta$), the update coefficient $\alpha = 1 - \rho/\eta$ became negative. Surprisingly, in this unsatisfied condition Halpern still converged whenever c was set smaller than 0.01.

2.2 Forsaken

We begin with the standard Forsaken problem

$$\min_x \max_y x(y - a) + f(x) - f(y),$$

where $f(z) = \frac{1}{4}z^2 - \frac{1}{2}z^4 + \frac{1}{6}z^6$.

In Example 6 of Pethick et al. [2], when $a = 0.45$ the parameter value is $\rho = 0.477761$ with candidate points $z' = \{-1.01236, -0.104749\}$. For this setting, with $\rho = 0.477761$ and $1/L = 0.08$, since $1/L < \rho$ violates the convergence condition, implying that a or boundary for the problem to reduce L must be tuned in order to apply Halpern successfully.

With the boundary condition fixed at $[-1.5, 1.5]$, we report below the values of ρ obtained for different choices of a , along with the corresponding candidate points (x, y) . The Lipschitz constant remains unchanged, so $1/L = 0.0806285$ for all cases.

Note that we report the absolute values of ρ .

Table 1: Forsaken parameters under fixed boundary $[-1.5, 1.5]$. Here $1/L = 0.0806285$ is constant, while ρ depends on a .

a	ρ	Candidate (x, y)	$1/L$	Satisfied?
1.00	0.660658	(0.1992, 0.4932)	0.0806285	×
0.65	0.621862	(−0.9949, −0.1013)	0.0806285	×
0.55	0.514417	(−1.0086, −0.0845)	0.0806285	×
0.50	0.486528	(−1.0116, −0.0889)	0.0806285	×
0.45	0.477761	(−1.0124, −0.1047)	0.0806285	×
0.40	0.479227	(−1.0119, −0.1272)	0.0806285	×
0.35	0.486469	(−1.0105, −0.1532)	0.0806285	×
0.25	0.509185	(−1.0057, −0.2090)	0.0806285	×
0.00	0.555246	(−0.9857, −0.3369)	0.0806285	×

As the next step, we modify the boundary condition to enforce the convergence requirement. Initially the boundary is set as $[-\frac{b}{2}, \frac{b}{2}]$ with $b = 3$. By reducing b we decrease L , while keeping $a = 0.45$ fixed. Table 2 reports the values of ρ (constant across all runs), $1/L$, and whether the convergence condition is satisfied. We observe that the condition first holds at $b = 2.2$, which will therefore be used in the subsequent experiments.

Table 2: Effect of boundary size b on the Forsaken problem with $a = 0.45$. The value of $\rho = 0.439773$ is constant across all runs, while $1/L$ depends on b .

b	ρ	$1/L$	Satisfied?
2.5	0.439773	0.267265	×
2.3	0.439773	0.433834	×
2.2	0.439773	0.505808	✓
2.1	0.439773	0.505808	✓

Having fixed the boundary at $b = 2.2$, where the convergence condition is satisfied, we next tune the parameter a to examine how ρ changes under this setting.

Table 3: Forsaken parameters with $b = 2.2$ fixed and varying a . Here $1/L = 0.505808$ is constant, while ρ and (x, y) vary with a .

a	ρ	Candidate (x, y)	$1/L$	Satisfied?
1.00	0.514325	$(-1.0022, -0.8939)$	0.505808	×
0.65	0.621862	$(-0.9949, -0.1013)$	0.505808	×
0.55	0.409233	$(0.9966, 0.4407)$	0.505808	✓
0.50	0.421854	$(0.9938, 0.4606)$	0.505808	✓
0.45	0.439773	$(0.9878, 0.4694)$	0.505808	✓
0.40	0.459849	$(0.9813, 0.4693)$	0.505808	✓
0.35	0.480423	$(0.9759, 0.4625)$	0.505808	✓
0.25	0.518074	$(0.9710, 0.4360)$	0.505808	×
0.00	0.555246	$(0.9857, 0.3369)$	0.505808	×

From Table 3, we observe that the convergence condition is satisfied only when a lies between 0.35 and 0.55. Among these, $a = 0.55$ yields the lowest value of ρ while still satisfying the condition, so we will primarily use $a = 0.55$ (together with $b = 2.2$) in the subsequent experiments.

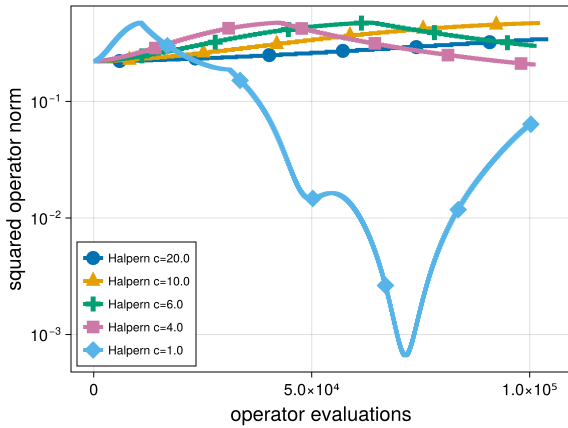
So the problem can now be written as

$$\min_x \max_y x(y - 0.55) + f(x) - f(y),$$

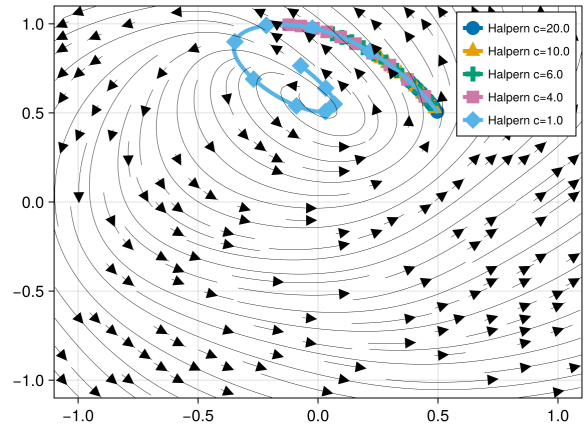
where $f(z) = \frac{1}{4}z^2 - \frac{1}{2}z^4 + \frac{1}{6}z^6$, with boundary $[-1.1, 1.1]$, where solution is $[-0.013107, 0.556549]$. In this setting we have $\rho = 0.409233$, and with $\eta = 0.85/L$ we obtain $\eta = 0.42994$, which satisfies the convergence condition.

We then compare Halpern runs with $c \in [1.0, 20.0]$ under two different iteration budgets.

For 10^5 iterations, only the run with $c = 1.0$ shows clear convergence, but even then it enters a limit cycle once the residual norm reaches around 10^{-3} . The other runs are noticeably slower.



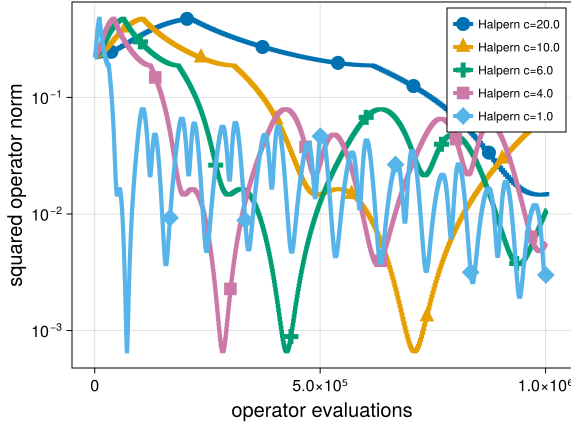
(a) Squared operator norm vs. operator evaluations



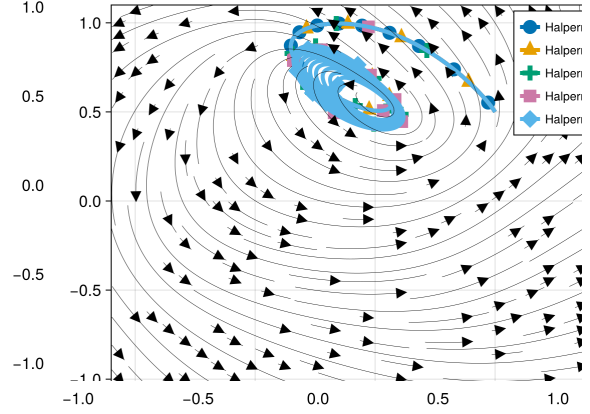
(b) Iterate trajectories over the vector field

Figure 5: Forsaken with $a = 0.55$ and boundary $[-1.1, 1.1]$ over 10^5 iterations. Settings: $\rho = 0.409233$, $\eta = 0.85/L = 0.42994$, $c \in [1.0, 20.0]$.

When the budget is extended to 10^6 iterations, all runs eventually converge, with $c = 1.0$ again showing the fastest initial decrease. Nevertheless, they still struggle to escape limit cycles once the residual norm drops to around 10^{-3} .



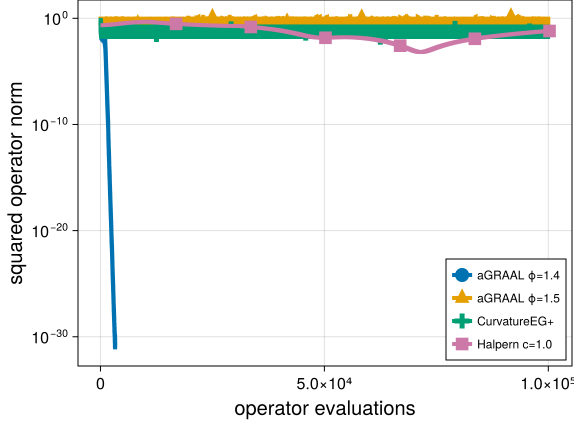
(a) Squared operator norm vs. operator evaluations



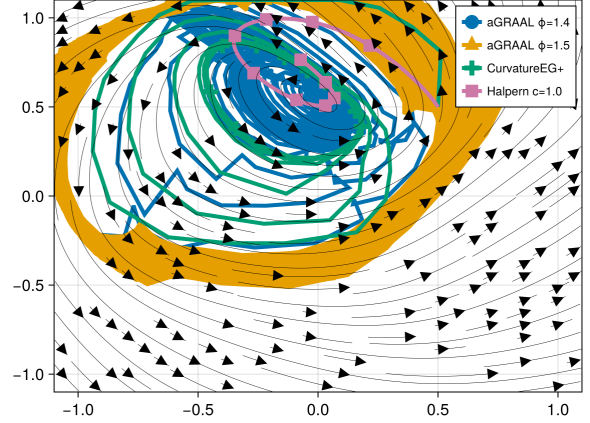
(b) Iterate trajectories over the vector field

Figure 6: Forsaken with $a = 0.55$ and boundary $[-1.1, 1.1]$ over 10^6 iterations. Settings: $\rho = 0.409233$, $\eta = 0.85/L = 0.42994$, $c \in [1.0, 20.0]$.

When comparing with other methods, we observe that all of them also struggle with limit cycles, except aGRAAL with $\phi = 1.4$, which converges reliably. We also note that under the broader step size conditions of Example 5.3.1 in Alacaoglu et al. [1] CurvatureEG+ was able to escape the limit cycle and converge, whereas our tuned($c=1.0$) Halpern did not.



(a) Squared operator norm vs. operator evaluations



(b) Iterate trajectories over the vector field

Figure 7: Comparison of Halpern with other methods on the Forsaken problem ($a = 0.55$, boundary $[-1.1, 1.1]$). Settings: $\rho = 0.409233$, $\eta = 0.85/L = 0.42994$. All methods exhibit limit-cycle behavior near residual norm 10^{-3} except aGRAAL with $\phi = 1.4$, which converges successfully.

In summary, Forsaken shows that once the convergence condition is enforced (by tuning $b = 2.2$ and $a = 0.55$), Halpern converges. However, it remains slow and exhibits limit-cycle behavior near residual norm 10^{-3} , while aGRAAL ($\phi = 1.4$) is the only method that escapes these cycles.

2.3 Polar Game

We follow Example 5.3.2 in Alacaoglu et al. [1]. For $a > 0$ the operator is

$$F(x, y) = (\psi(x, y) - y, \psi(y, x) + x), \quad \psi(x, y) = \frac{a}{4} x(-1 + x^2 + y^2)(-1 + 4x^2 + 4y^2).$$

Case $a = \frac{1}{3}$ For this setting we obtain $\rho = 0.093544$ and $1/L = 0.143999$, so the convergence condition $\rho < 1/L$ is satisfied. With $\eta = 0.8/L \approx 0.115200$, Halpern converges smoothly. All tested c values in the range $[0.01, 20]$ converge, with $c = 0.01$ giving the best performance.

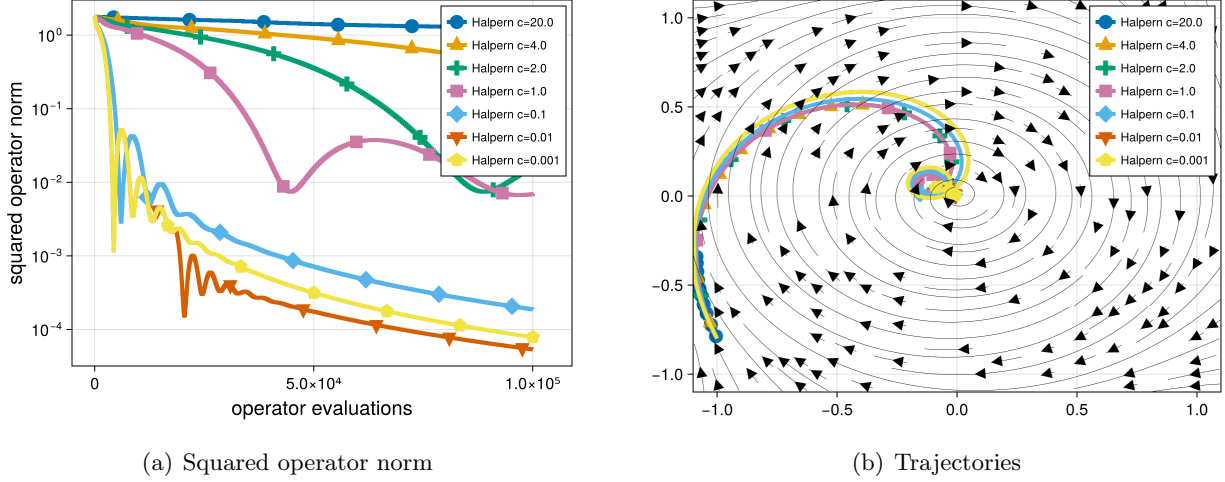


Figure 8: Polar Game ($a = \frac{1}{3}$). $\rho = 0.093544$, $1/L = 0.143999$, $\eta = 0.8/L = 0.115200$. Condition satisfied; all runs with $c \in [0.01, 20]$ converge, best at $c = 0.01$.

However, compared with aGRAAL ($\phi = 1.3$), Halpern is much slower; aGRAAL achieves a rapid decrease, while Halpern requires orders of magnitude more iterations.

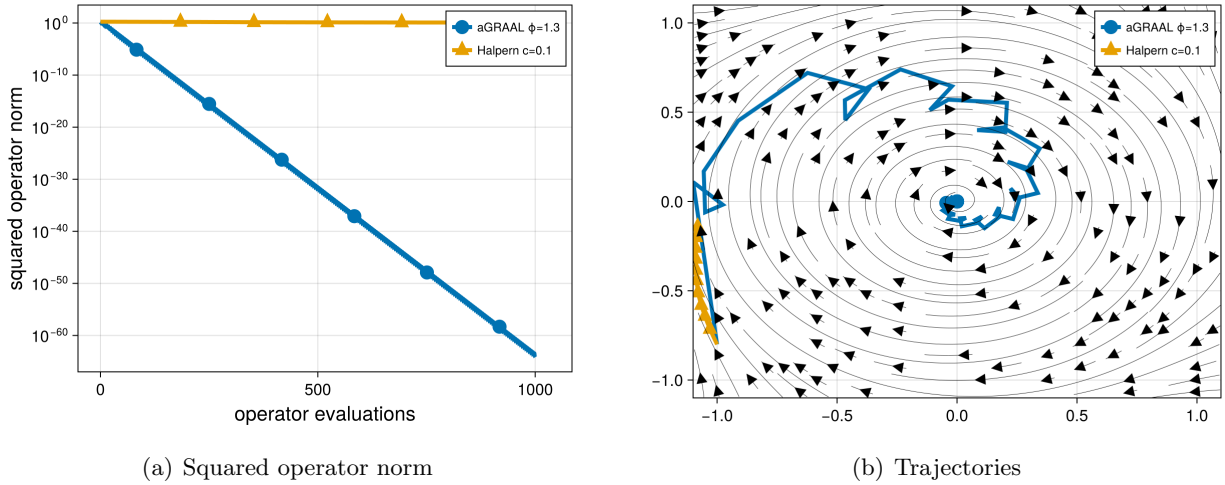


Figure 9: Polar Game ($a = \frac{1}{3}$). aGRAAL ($\phi = 1.3$) vs. Halpern. Both converge, but Halpern is much slower.

Case $a = 3$ Here we have $\rho = 0.716269$ and $1/L = 0.016291$, so $\rho < 1/L$ is violated. With the given ρ and $\eta = 0.8/L \approx 0.013032$, Halpern diverges.

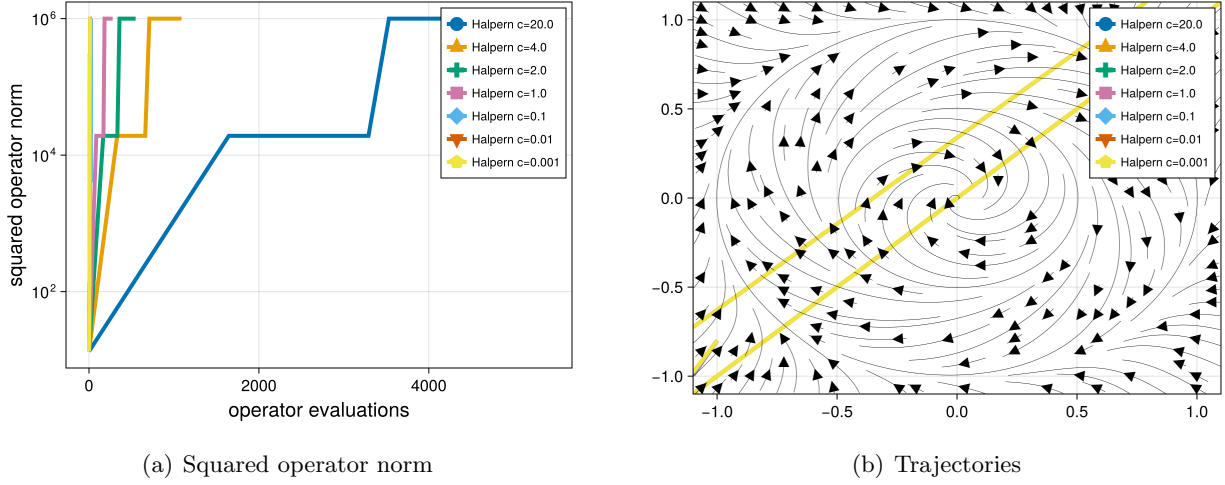


Figure 10: Polar Game ($a = 3$). Given $\rho = 0.716269$ and $1/L = 0.016291$, the condition fails and Halpern diverges (with $\eta = 0.8/L = 0.013032$).

We then damp the algorithm by setting $\rho = 0.25 \eta$ with $\eta = 0.8/L = 0.013032$, so $\rho = 0.003258$. This satisfies $0 < \rho < \eta < 1/L$. The residual now decreases initially, but the run settles into a limit cycle; among the tested values $c \in [0.01, 20]$, $c = 1.0$ performs best.

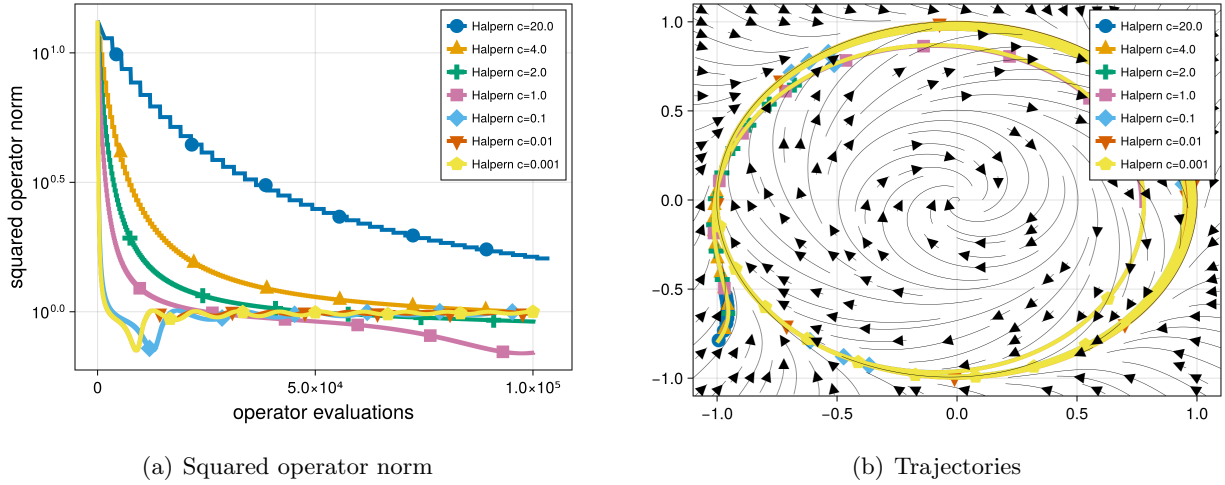


Figure 11: Polar Game ($a = 3$). With $\eta = 0.8/L = 0.013032$ and $\rho = 0.25 \eta = 0.003258$, Halpern improves initially but gets trapped in a limit cycle; $c = 1.0$ is best among $c \in [0.01, 20]$.

Finally, compared to aGRAAL ($\phi = 1.3$), Halpern ($c = 1.0$) remains in a limit cycle even beyond 10^6 iterations.

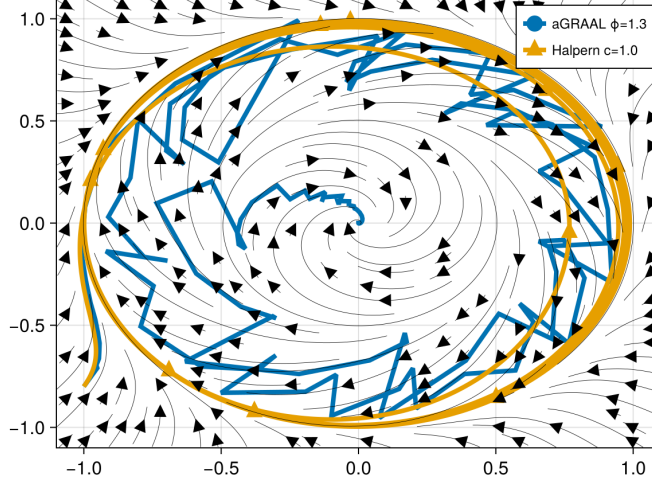


Figure 12: Polar Game ($a = 3$). Halpern ($c = 1.0$) vs. aGRAAL ($\phi = 1.3$): Halpern remains in a limit cycle after 10^6 iterations.

Overall, in the Polar Game, Halpern converges only in the easy case ($a = \frac{1}{3}$) and remains much slower than aGRAAL, while in the hard case ($a = 3$) it either diverges or gets stuck in a limit cycle.

3 Conclusion

Across the three benchmarks, the inexact Halpern method shows a clear pattern of sensitivity to parameter choices.

On the **Lower Bound** instance, convergence was possible only when the condition $\rho < \eta < 1/L$ was satisfied, or otherwise with extremely small c values.

For **Forsaken**, enforcing the condition by tuning both a and the boundary size allowed convergence, but the method remained slow and repeatedly entered limit cycles once the residual norm reached 10^{-3} . Among the methods compared, only aGRAAL with $\phi = 1.4$ escaped the cycles under our tuned condition, unlike in the original setting.

In the **Polar Game**, Halpern converged slowly for the easier case ($a = \frac{1}{3}$) but completely stalled for the harder case ($a = 3$), where ρ exceeded $1/L$ and the condition was violated.

Overall, Halpern converges once the condition is enforced, but the behavior is fragile method depends heavily on careful tuning of η , ρ , and c , and in harder problems it is clearly less reliable than more robust methods such as aGRAAL.

References

- [1] Ahmet Alacaoglu, Axel Böhm, and Yura Malitsky. Beyond the golden ratio for variational inequality algorithms, 2022.
- [2] Thomas Pethick, Puya Latafat, Panagiotis Patrinos, Olivier Fercoq, and Volkan Cevher. Escaping limit cycles: Global convergence for constrained nonconvex-nonconcave minimax problems, 2023.